
Statement of Technical Capabilities

in Electromagnetics and
Electromechanical Control

Prepared for the
Immersive Hiking Simulator Treadmill Project

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1. Executive Summary

This document presents the technical capabilities of Shanghai Qiongpei Technology Co., Ltd. as they relate to the Immersive Hiking Simulator Treadmill Project. Our assessment is grounded in five production-validated technology platforms that directly address the core engineering challenges of this project.

Core Argument

The central challenge of the Immersive Hiking Simulator Treadmill is not the selection of any single component. It is the **system-level integration** of electromagnetic arrays, drive control, multi-sensor feedback, haptic response, audio-visual synchronization, and safety mechanisms into a reliable, maintainable engineering system.

This class of problem—multi-actuator, multi-sensor, real-time closed-loop control with stringent safety requirements—is precisely what our team has been building and shipping for the past three years.

Our existing technology capabilities map directly to the project requirements:

Capability Area	What We Have Built	Project Relevance
Motor Drive & Control	Self-developed FOC drive boards, current regulation, thermal protection	Electromagnetic array drive and protection
Multi-Actuator Systems	16-DOF robotic platform with multi-sensor fusion	Multi-channel array coordination
Real-Time Control	FSM architecture, gait control, dynamic state switching	Terrain-adaptive feedback control
Safety & Fault Handling	E-Stop, safety gates, fault recovery, real-time telemetry	System safety and fault tolerance

2. Existing Technical Capabilities

2.1 Electromagnetic Drive and Low-Level Control

Our self-developed motor drive platform provides production-validated capabilities in electromagnetic actuation and precision current control.

Hardware Platform:

- High-performance Field-Oriented Control (FOC) on STM32G4 microcontroller

- CAN bus communication interface for multi-device coordination
- Triple closed-loop control: position, velocity, and torque
- High-precision current sampling with offset calibration
- Hardware-level overtemperature protection and fault shutdown logic

Demonstrated Competencies:

- Electromagnetic actuator drive circuit design and validation
- Precision current regulation and power envelope management
- High-frequency PWM control with sub-millisecond response latency
- Thermal modeling, protection threshold definition, and derating strategy
- Coordinated design across drive boards, control firmware, and host systems

Relevance to Electromagnetic Array Control

Although the project employs electromagnetic arrays rather than rotary motors, the two are closely analogous at the drive-and-control level. Both involve coil excitation waveform design, precision current regulation, high-frequency switching response, thermal boundary management, and fault protection. Our drive platform addresses all of these requirements with proven, production-grade firmware.

2.2 Multi-Actuator, Multi-Sensor Integrated Systems

Our industrial wheeled-legged quadruped robot platform is a multi-actuator, multi-sensor, real-time closed-loop control system that has achieved stable locomotion across six terrain types.

System Specifications:

- Wheeled-legged quadruped configuration with 16 degrees of freedom
- Self-developed motor drive and actuation system
- Multi-modal sensor suite: LiDAR, depth cameras, IMU, joint encoders
- Unified control architecture spanning motion control, state machines, safety logic, and supervisory interfaces
- Demonstrated stable traversal across six terrain types (flat, slope, stairs, gravel, grass, uneven ground)

Demonstrated Competencies:

- Parallel coordination of 16+ actuators with synchronized timing
- Multi-sensor data fusion with deterministic timing guarantees
- Real-time closed-loop control at 50 Hz with sub-2 ms inference latency
- System-level integration, commissioning, and stability validation
- Terrain-differentiated control strategy design and switching

Relevance to Multi-Channel Array Control

The electromagnetic treadmill array is fundamentally a multi-actuator system requiring spatially distributed, individually addressable control—the same class of problem as coordinating 16 independent motor drives on a quadruped platform. Our demonstrated capability to maintain stable, synchronized control across multiple actuators under dynamic conditions transfers directly to array-level electromagnetic control.

2.3 Real-Time Motion Control and State Machine Architecture

Our motion control stack provides a production-grade framework for real-time state management and dynamic behavior switching.

Architecture:

- Finite State Machine (FSM) with formally defined states and safe transitions
- Remote command interface with priority-based control arbitration
- 50 Hz control loop with real-time inference and PD gain scheduling
- Modular architecture supporting third-party integration via standard APIs

Demonstrated Competencies:

- Design and implementation of multi-state control logic with safe transitions
- Real-time dynamic parameter switching (PD gains, motion profiles, terrain modes)
- Coordination between control dispatch, command arbitration, and safety interlocks
- SDK-level API design for system integration by third-party developers

Relevance to Dynamic Terrain Feedback

Electromagnetic array control for a hiking simulator is not a static output problem. It is a dynamic system that must continuously adapt to footstep events, walking speed, terrain mode selection, and safety status. This is precisely the class of problem that state machines and real-time control frameworks are designed to solve. Our control architecture can be directly adapted to manage terrain-mode switching, cadence-responsive feedback, and multi-modal synchronization.

2.4 Safety Control, Fault Handling, and Remote Maintenance

Our runtime safety framework provides comprehensive protection and fault management for electromechanical systems.

Safety Architecture:

- Hardware E-Stop safety layer integrated into the system architecture
- Safety gates: every actuation command passes through capability checks before execution
- Structured fault detection, classification, and recovery logic
- Real-time telemetry for all critical system parameters
- End-to-end auditability with timestamped event logs

Direct Correspondence to Project Requirements:

Project Safety Requirement	Existing Capability
Electromagnetic array anomaly protection	Hardware fault detection + automatic shut-down
Overtemperature protection	Thermal monitoring + derating + E-Stop
Emergency stop logic	Hardware E-Stop + software safety gates
Sensor fault handling	Structured fault classification + recovery
Long-duration operational stability	Telemetry + thermal management + watch-dogs
Ongoing firmware maintenance	Remote update capability with rollback support

3. Direct Mapping to Project Requirements

The Immersive Hiking Simulator Treadmill Project presents six core technical challenges. Each maps to specific, validated capabilities within our technology stack.

Project Challenge	Mapped Capability
1. Stable, adjustable resistance feedback from electromagnetic array and copper woven conductive layer	Precision current regulation, closed-loop torque control, thermal protection
2. Spatially distributed multi-channel control rather than single constant output	Parallel multi-actuator coordination with independent per-channel control
3. Dynamic feedback triggered by footsteps, speed variation, and terrain mode	FSM-based state switching, real-time event-driven control, cadence detection
4. Synchronization of electromagnetic feedback with haptics, audio, video, and safety	Deterministic timing architecture, multi-subsystem coordination, priority arbitration
5. Long-duration stability with controlled temperature rise and abnormal-condition handling	Thermal management framework + safety layer + real-time telemetry monitoring
6. Simulation-driven design convergence for magnetic field distribution, coupling, and thermal boundaries	Electromagnetic simulation methodology + parametric validation workflow

Engineering Match, Not Conceptual Match

The correspondence between our capabilities and this project’s requirements is not theoretical. It is grounded in **production-validated** systems: drives that have powered real actuators, control stacks that have maintained stability across real terrain, safety systems that have protected real hardware, and maintenance systems that have updated real devices in the field.

4. Professional Assessment of Project Challenges

From an engineering-implementation standpoint, the most difficult aspects of this project are not in the electromagnetic array itself, but in the following system-level integration challenges:

1. **Subjective feel calibration.** Tuning the damping and resistance feedback to feel natural and immersive requires iterative testing with human subjects. This cannot be solved through

simulation alone—it requires rapid prototyping capability and a control system that supports real-time parameter adjustment.

2. **Terrain differentiation fidelity.** Different terrain types (rock trail, forest path, sand, snow) must produce perceptibly distinct feedback profiles. This requires a control architecture that supports multiple terrain modes with smooth transitions between them.
3. **Cadence consistency.** The feedback must remain consistent under varying left-right foot triggering patterns and walking speeds. This is a multi-sensor fusion and timing-coordination problem.
4. **Multi-modal synchronization.** Electromagnetic resistance feedback must be precisely synchronized with haptic actuators, spatial audio, and visual projection. Latency mismatches as small as 50 ms degrade the immersive experience.
5. **Thermal management under sustained operation.** Commercial installations may operate 8–12 hours per day. Thermal management and stability must be engineered from the ground up, not added as an afterthought.
6. **Retrofit-space constraints.** Integrating electromagnetic, mechanical, electrical, and control systems within the physical constraints of a treadmill form factor requires coordinated multi-discipline design.

The common characteristic of these challenges is that they must be resolved through **progressive validation and de-risking**—combining electromechanical control experience, real-time systems experience, and system-level integration capability. Theoretical analysis alone is insufficient. Our team’s capabilities have been built around precisely this class of system-level engineering problem.

5. Conclusion

Shanghai Qiongpei Technology Co., Ltd. possesses the following tangible, production-validated technical capabilities that are directly relevant to the Immersive Hiking Simulator Treadmill Project:

1. **Self-developed FOC drive and low-level electromagnetic control** — precision current regulation, triple closed-loop control, and thermal protection on custom hardware
2. **Multi-actuator, multi-sensor robotic system development** — 16-DOF parallel actuation, multi-modal sensing, and terrain-adaptive control

3. **Real-time motion control and state machine architecture** — FSM-based control, dynamic parameter switching, and SDK-level integration interface
4. **Safety control and fault handling** — E-Stop safety layer, safety gates, structured fault recovery, and real-time telemetry
5. **Electromagnetic simulation and parametric design convergence** — Methodology for magnetic-field distribution analysis, thermal-boundary verification, and prototype parameter definition
6. **System-level integration and commissioning** — Demonstrated ability to bring multi-discipline systems from prototype to stable operation

Summary

On the basis of these capabilities, our team is well positioned to undertake the electromagnetic system design, drive control development, sensing closed-loop implementation, system integration, and commissioning and validation work required for the prototype phase of this project.

We welcome the opportunity to discuss the project in further detail and to define a concrete engagement plan.

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